

EDUCATION

Swarthmore College — Philadelphia, PA

Aug 2023—May 2027

B.S. General Engineering (Electrical focus) & B.A. Applied Mathematics | GPA: 3.98/4.00

Relevant coursework: Electronic Circuit Applications, Digital Signal Processing, Computer Vision, Data Structures & Algorithms (C++), Ordinary & Partial Differential Equations (PDEs), Real Analysis, Tensor Decompositions, Numerical PDEs, Thermo-Fluid Mechanics, Electromagnetism, Electrodynamics, Statistical Signal Processing

Teaching Assistant for Electrical Circuit Analysis (ENGR 011), Aug 2025—Jan 2026

NCAA Varsity Soccer, Aug 2023—Jun 2026

RESEARCH AND WORK EXPERIENCE

Advanced Materials Intern & Research Assistant – US Naval Research Laboratory, Washington, DC

May 2025—Aug 2025

- Measured thermal diffusivity and specific heat of thermoelectric semiconductors using laser flash analysis (LFA).
- Built Arduino-controlled LN₂ dispenser with closed-loop thermocouple monitoring, automating LFA experiments
- Created graphical-user-interface and data-logging interfacing between LN₂ dispenser and lab computer.
- Developed experiments to improve the lab’s measurement procedure for IR-transparent and high-resistance samples.
- Poster presentations: Swarthmore *Sigma Xi* 2025 poster session.

Applied Mathematics Research Assistant – Swarthmore College Mathematics, Philadelphia, PA

Jan 2024—present

- Utilizing computational fluid dynamics and numerical methods to research high-order accurate PDE solvers for kinetic & fluid plasma models in MATLAB.
- Developed low-rank, structure-preserving solver for Vlasov-Fokker-Planck kinetic-ion, fluid-electron plasma model.
- [Paper](#) submitted Aug 2026.
- Poster presentations: 2025 SIAM NNP conference, 2024 & 2025 *Sigma Xi* Swarthmore poster sessions.

Electrical Engineering Research Assistant -- Swarthmore College Engineering, Philadelphia, PA

Nov 2023—May 2024

- Researched wind-energy harvesting technology and used Arduino, MATLAB & ViscousFlow to simulate air vortex-shedding & electrical induction power output for an oscillatory wind-energy harvester.

Software Engineering Summer Intern – Oregon Health and Science University, Portland, OR

Jun 2022—Aug 2022

- Used Kotlin to develop Android app that detects heart murmurs using audio data from a Bluetooth stethoscope.
- Presented machine-learning paper to the lab’s reading group.

PROJECTS

Convolutional Neural Network for Hand-Drawn Image Classification: [project link](#)

May 2026

- Designed and trained a CNN to classify 250 object categories, deployed it online for real-time evaluation using ONNX.

Analog Circuit and PCB Design – Fusion360

Aug 2025—Dec 2025

- Designed and fabricated PCBs combining MOSFETs, sensors, actuators, op-amps, comparators, and audio amplifiers across four projects: [audio equalizer](#), [temp-controlled fan](#), [light-optimizing solar panel](#), [regulated DC power supply](#).

AI Python Stock Trading: trained neural network on technical indicators to trade stocks using Alpaca API, [link](#)

Feb 2025

Generative Adversarial Network (GAN): Image-generating model trained on web-scraped art, [link](#)

Apr 2022—Feb 2023

Technical & Professional Skills

Technical: SPICE, PCB Design, CAD, Fusion360, Soldering, Embedded systems, PDE solvers, Numerical methods

Programming: Python, MATLAB, Arduino, Java, C++, HTML/JavaScript, LaTeX

Software: VSCode, Git, SolidWorks, AutoCAD, NumPy, TensorFlow, PyTorch

Languages: Spanish (Fluent), Global Seal of Biliteracy (2022) | Citizenship: USA, Ireland

Honors and Scholarships

Tau Beta Pi Engineering Honors Society

Mar 2026

Delaware Valley Engineers Undergraduate Scholarship, Delaware Valley Engineers Society

Feb 2025, Mar 2026

Donna Prentice Memorial Scholarship, American Society of Civil Engineers

Feb 2024

National Merit Scholarship, National Merit Scholarship Corporation

Apr 2023